0.1 Butterfly Lemma

Lemma 0.1.1. Let G be a group, and let $H, K \leq G$ with $H^* \trianglelefteq H$ and $K^* \trianglelefteq K$. Then the following statements hold:

1) $H^*(H \cap K^*) \trianglelefteq H^*(H \cap K)$
2) $K^*(H^* \cap K) \trianglelefteq K^*(H \cap K)$
3) $\dfrac{H^*(H \cap K)}{H^*(H \cap K^*)} \cong \dfrac{K^*(H \cap K)}{K^*(H^* \cap K)} \cong \dfrac{H \cap K}{(H^* \cap K)(H \cap K^*)}$

Proof. Since $H^* \trianglelefteq H$, for any $h \in H \cap K$, we have $hH^* = H^*h$. Note that

$$H^*(H \cap K^*) = \bigcup_{h \in H \cap K^*} hH^* = \bigcup_{h \in H \cap K^*} H^*h = (H \cap K^*)H^*$$

Thus, $H^*(H \cap K^*) \leq H^*(H \cap K)$. Meanwhile, to show $H \cap K^* \trianglelefteq H \cap K$, let $x \in H \cap K$. Then

$$x(H \cap K^*)x^{-1} = (xHx^{-1}) \cap (xK^*x^{-1}) = H \cap K^*$$

Similarly, $H^* \cap K \trianglelefteq H \cap K$. Since the product of two normal subgroups is normal, we define:

$$L = (H \cap K^*)(H^* \cap K) \trianglelefteq H \cap K.$$

Define a map $\varphi : H^*(H \cap K) \to (H \cap K)/L$ by:

$$\varphi(hx) = xL \quad (h \in H^*, x \in H \cap K).$$

Well-defined: For $h_1, h_2 \in H^*$ and $x_1, x_2 \in H \cap K$,

$$h_1x_1 = h_2x_2 \implies h_2^{-1}h_1 = x_2x_1^{-1} \in H^* \cap (H \cap K) = H^* \cap K \leq L.$$

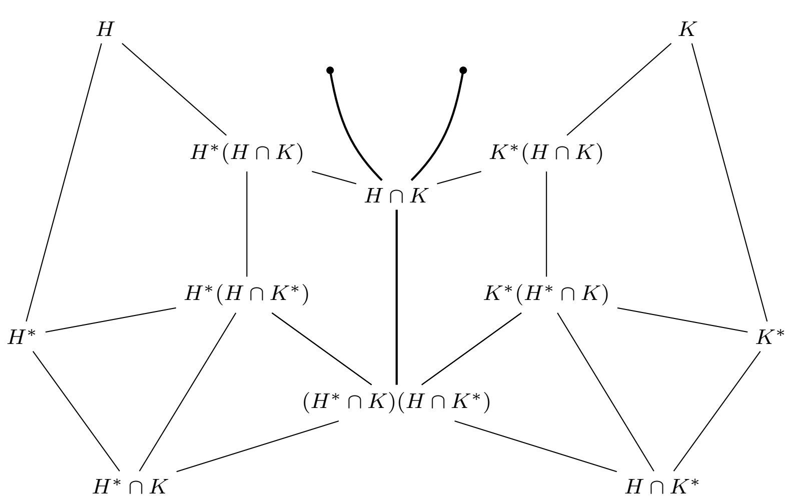
Thus, $x_2x_1^{-1} \in L \iff x_1L = x_2L$.
Homomorphism: For $h_1, h_2 \in H^*$ and $x_1, x_2 \in H \cap K$,

$$\begin{aligned}\varphi((h_1x_1)(h_2x_2)) &= \varphi(h_1(x_1h_2x_1^{-1})x_1x_2) \\ &= \varphi(h_1h_3x_1x_2) \quad (\text{for some } h_3 \in H^* \text{ since } H^* \trianglelefteq H) \\ &= (x_1x_2)L = (x_1L)(x_2L) = \varphi(h_1x_1)\varphi(h_2x_2).\end{aligned}$$

Onto: This is trivial as any $xL \in (H \cap K)/L$ has a pre-image $1 \cdot x \in H^*(H \cap K)$.
Kernel:

$$\begin{aligned}\ker\varphi &= \{hx \mid h \in H^*, x \in H \cap K, x \in L\} \\ &= \{hx \mid h \in H^*, x \in L\} \\ &= H^*L = H^*(H \cap K^*)(H^* \cap K) \\ &= H^*(H \cap K^*)\end{aligned}$$

Now, by the First Isomorphism Theorem, we obtain the result. □



0.2 Subnormal Series of Group

Definition 0.2.1. Let G be a Group.
A Chain of subsets of G $\{A_i\}_{i=0}^n$ is called *subgroup series* if:

$$1 = A_0 \leq A_1 \leq \cdots \leq A_{n-1} \leq A_n = G$$

Definition 0.2.2. Let G be a Group. A subgroup series $\{A_i\}_{i=0}^n$ is called *subnormal series* of G if:

$$1 = A_0 \trianglelefteq A_1 \trianglelefteq \cdots \trianglelefteq A_{n-1} \trianglelefteq A_n = G$$

And for each $0 \leq i \leq n-1$, A_{i+1}/A_i is called i^{th} *factor group*.
Specially, if for all $0 \leq i \leq n-1$, $A_i \trianglelefteq G$, then it is called *normal series*.

0.3 Schreier Refinement Theorem

Theorem 0.3.1. Any two subnormal series of a group G have isomorphic refinements.

Proof. Let G be a group, and let

$$\{H_i\}_{i=0}^n : 1 = H_0 \trianglelefteq H_1 \trianglelefteq \cdots \trianglelefteq H_n = G$$
$$\{K_j\}_{j=0}^m : 1 = K_0 \trianglelefteq K_1 \trianglelefteq \cdots \trianglelefteq K_m = G$$

be two subnormal series for G. For each $0 \leq i \leq n-1$ and $0 \leq j \leq m$, define:

$$H_{i,j} \overset{\text{def}}{=} H_i(H_{i+1} \cap K_j).$$

Then, $H_{i,j} \trianglelefteq H_{i,j+1}$ for each j because

$$H_{i,j} = H_i(H_{i+1} \cap K_j) \trianglelefteq H_i(H_{i+1} \cap K_{j+1}) = H_{i,j+1}.$$

(This follows from the fact that $K_j \trianglelefteq K_{j+1} \implies H_{i+1} \cap K_j \trianglelefteq H_{i+1} \cap K_{j+1}$).
In the same sense, for each $0 \leq j \leq m-1$ and $0 \leq i \leq n$, we define:

$$K_{j,i} \overset{\text{def}}{=} K_j(K_{j+1} \cap H_i).$$

Now, we can construct refinements for each subnormal series as follows:

$$\begin{aligned}
H_0 &= H_{0,0} \trianglelefteq H_{0,1} \trianglelefteq \cdots \trianglelefteq H_{0,m-1} \trianglelefteq H_{0,m} = H_1 \\
H_1 &= H_{1,0} \trianglelefteq H_{1,1} \trianglelefteq \cdots \trianglelefteq H_{1,m-1} \trianglelefteq H_{1,m} = H_2 \\
&\vdots \\
H_{n-1} &= H_{n-1,0} \trianglelefteq H_{n-1,1} \trianglelefteq \cdots \trianglelefteq H_{n-1,m} = G
\end{aligned}$$

$$\begin{aligned}
K_0 &= K_{0,0} \trianglelefteq K_{0,1} \trianglelefteq \cdots \trianglelefteq K_{0,n-1} \trianglelefteq K_{0,n} = K_1 \\
K_1 &= K_{1,0} \trianglelefteq K_{1,1} \trianglelefteq \cdots \trianglelefteq K_{1,n-1} \trianglelefteq K_{1,n} = K_2 \\
&\vdots \\
K_{m-1} &= K_{m-1,0} \trianglelefteq K_{m-1,1} \trianglelefteq \cdots \trianglelefteq K_{m-1,n} = G
\end{aligned}$$

Moreover, the **Butterfly Lemma** gives the following isomorphism: for all $0 \leq i \leq n-1$ and $0 \leq j \leq m-1$,

$$H_{i,j+1}/H_{i,j} = \frac{H_i(H_{i+1} \cap K_{j+1})}{H_i(H_{i+1} \cap K_j)} \cong \frac{K_j(K_{j+1} \cap H_{i+1})}{K_j(K_{j+1} \cap H_i)} = K_{j,i+1}/K_{j,i}.$$

In summary, the following matrices are Isomorphic elementwise. (Look at the Beautiful Symmetric Relation!!)

$$\begin{bmatrix}
H_{0,1}/H_{0,0} & H_{0,2}/H_{0,1} & \cdots & H_{0,m}/H_{0,m-1} \\
H_{1,1}/H_{1,0} & H_{1,2}/H_{1,1} & \cdots & H_{1,m}/H_{1,m-1} \\
\vdots & \vdots & \ddots & \vdots \\
H_{n-1,1}/H_{n-1,0} & H_{n-1,2}/H_{n-1,1} & \cdots & H_{n-1,m}/H_{n-1,m-1}
\end{bmatrix} \cong \begin{bmatrix}
K_{0,1}/K_{0,0} & K_{0,2}/K_{0,1} & \cdots & K_{0,n}/H_{0,n-1} \\
K_{1,1}/K_{1,0} & K_{1,2}/K_{1,1} & \cdots & K_{1,n}/H_{1,n-1} \\
\vdots & \vdots & \ddots & \vdots \\
K_{m-1,1}/K_{m-1,0} & K_{m-1,2}/K_{m-1,1} & \cdots & K_{m-1,n}/H_{m-1,n-1}
\end{bmatrix}^T$$

Consequently, the two subnormal series have isomorphic refinements. ∎

0.4 Jordan-Hölder Theorem

Lemma 0.4.1. Every non-trivial finite group has a maximal normal subgroup.

Proof. Let G be a non-trivial finite group. If G is a simple group, then its only proper normal subgroup is the trivial group $\{1\}$. Thus, $\{1\}$ is the maximal normal subgroup of G, and the proof is complete.
Suppose G is not simple. Then by definition, G has at least one non-trivial proper normal subgroup N.
If N is a maximal normal subgroup of G, there is nothing to prove.
If N is not a maximal normal subgroup, then by definition, there exists a proper normal subgroup N_1 of G such that

$$N < N_1 \lhd G$$

If N_1 is not maximal, we can continue this process to obtain a strictly increasing chain of proper normal subgroups:

$$N < N_1 < N_2 < \cdots \lhd G$$

Since G is a finite group, the order of these subgroups is bounded by the order of G (i.e., $|N_i| < |G|$). Because the size of the subgroups must strictly increase at each step $(|N| < |N_1| < |N_2| < \dots)$, this process must terminate in a finite number of steps.
The last subgroup in this terminating chain will be a proper normal subgroup M such that there is no other normal subgroup between M and G. Therefore, G has a maximal normal subgroup. □

Definition 0.4.2. Let G be a group. A subnormal series $\{N_i\}_{i=0}^n$ is called *composition series* if:

For all $0 \leq i \leq n-1$, N_{i+1}/N_i is a simple group

Theorem 0.4.3. Let G be a Finite Group. Then, the follwing statements hold:

1. G has a Composition Series.

2. Every composition series of G is unique.

More precisely, if $\{N_i\}_{i=0}^n$ and $\{M_i\}_{i=0}^m$ are two composition series for G, then $n = m$ and there exists a permutation π of $\{1, 2, \dots, n\}$ such that

$$M_{\pi(i)}/M_{\pi(i)-1} \cong N_i/N_{i-1}$$

In sum,

Every Finite Group has a unique Composition Series

Proof. By Lemma, statement 1. is proved:
G has a Maximal normal subgroup $N_0 \lhd G$, and N_0 has a Maximal normal subgroup $N_1 \lhd N_0$, this process must terminate in a finite number of steps.
Moreover, every composition series has no refinement when duplications are ignored, and thus the Schreier Refinement Theorem ensures uniqueness. □

Definition 0.4.4. A Group G is called *solvable* if: There exists a subnormal series $\{H_i\}_{i=0}^{n}$ such that

For all $0 \le i \le n-1$, H_{i+1}/H_i is a Abelian Group

Proposition 0.4.5. If G is an Abelian simple group, then $G \cong \mathbb{Z}_p$ for some prime p. (The converse is also true.)

Proof. Since G is Abelian, every proper non-trivial subgroup $1 \lneq H \lneq G$ is normal. The simplicity of G implies that G has no proper non-trivial subgroup. That is, for any non-zero $g \in G$, we must have $\langle g \rangle = G$.
Now, $G \cong \mathbb{Z}_k$ for some $k \in \mathbb{N}$ or $G \cong \mathbb{Z}$. If $G \cong \mathbb{Z}$, then it has $2\mathbb{Z}$ as a proper non-trivial subgroup, which contradicts the simplicity of G. Thus, $G \cong \mathbb{Z}_n$ for some $n \in \mathbb{N}$. If n is not prime, say $n = mk$ with $1 < m < n$, then it has a proper non-trivial subgroup $m\mathbb{Z}_n$, which again contradicts simplicity. Therefore, $G \cong \mathbb{Z}_p$ for some prime p. □

Lemma 0.4.6. Let G be a group and $N \trianglelefteq G$ be a normal subgroup. Then,

G/N is simple *if and only if* N is a maximal normal subgroup of G.

Proof. By the 4th Isomorphism Theorem, we have:

$$\begin{aligned} G/N \text{ is simple } &\iff G/N \text{ has no proper non-trivial normal subgroup} \\ &\iff \text{There is no normal subgroup } H \text{ such that } N \lhd H \lhd G \\ &\iff N \text{ is a maximal normal subgroup of } G \end{aligned}$$

Proposition 0.4.7. Let G be a finite group. Then,

G is solvable *if and only if* G has a composition series such that all its factors are Abelian

Proof. $(\Rightarrow)$ Suppose that G is solvable. That is, G has a subnormal series $\{H_i\}_{i=0}^{n}$ such that

$$1 = H_0 \lhd H_1 \lhd \cdots \lhd H_{n-1} \lhd H_n = G$$

where each factor H_{i+1}/H_i is Abelian.
For each H_{i+1}/H_i $(0 \le i \le n-1)$, since it is a finite Abelian group, it has a composition series $\{H_{i,j}/H_i\}_{j=0}^{r_i}$ such that

$$1 = H_i/H_i = H_{i,0}/H_i \lhd H_{i,1}/H_i \lhd \cdots \lhd H_{i,r_i}/H_i = H_{i+1}/H_i.$$

by the Jordan-Hölder Theorem. That is, each factors

$$(H_{i,j+1}/H_i)/(H_{i,j}/H_i)$$

are simple. Since H_{i+1}/H_i is Abelian, these factors are also Abelian.
The 4th Isomorphism Theorem gives a series $\{H_{i,j}\}_{j=0}^{r_i}$ such that

$$H_i = H_{i,0} \lhd H_{i,1} \lhd \cdots \lhd H_{i,r_i} = H_{i+1}$$

Indeed, all factors of this series are simple and Abelian because the 3rd Isomorphism Theorem gives:

$$(H_{i,j+1}/H_i)/(H_{i,j}/H_i) \cong H_{i,j+1}/H_{i,j}$$

Consequently, by concatenating these series, we obtain

$$1 = H_0 = H_{0,0} \lhd H_{0,1} \lhd \cdots \lhd H_{0,n_0} = H_1 \lhd H_{1,1} \lhd \cdots \lhd H_{n-1,r_{n-1}} = H_n = G$$

which is a composition series satisfying the condition that every factor is Abelian. The converse is trivial. □

Corollary 0.4.8. Let G be a finite group. Then,

G is solvable *if and only if* G has a composition series such that all its factors are cyclic of prime order.